

Draft provisional materials disclosure

Prepared 29 September 2026 | Technical working draft for applicant and patent counsel

This draft develops the requested all-element composition framework, including single-element materials, every component count from 2 through 138, exact ranges, isotope parameters, particle categories, and conditional property scenarios. The numbered description and candidate claim annex are intended to be reviewed together. No application has been filed. No applicant laboratory results, identified inventive contribution, or full-scope enablement determination has been supplied. Mathematical definitions are exact; they are not evidence that every defined material can be prepared.

1 Technical field

[0001] The subject matter concerns the description, preparation, characterization, and comparison of materials having specified elemental and isotopic compositions, structures, processing histories, and properties. Materials considered include metals and alloys, inorganic compounds and ceramics, molecular substances, polymers, composites, mixtures, and single-element forms. Particle-containing or excited systems are addressed separately from ordinary chemical constituents.

[0002] A composition framework is provided to identify selected constituents, normalize their proportions, express exact values and intervals, and associate each numerical description with a structural and preparation record. Its purpose is to make the requested scope explicit and reproducible as a specification of candidate compositions. Whether an individual candidate exists, can be synthesized, or has a useful property requires additional technical information.

2 Background and evidence status

[0003] An elemental formula does not uniquely define a material. Connectivity, crystal structure, phase assemblage, isotope distribution, defects, interfaces, orientation, morphology, and processing can distinguish systems sharing the same elemental proportions. A molecular formula can describe different isomers; a ceramic formula can describe different polymorphs; a polymer formula can omit sequence and architecture; a composite's overall percentages can omit its load-bearing or conducting network.

[0004] The element vocabulary has two groups. The recognized group consists of atomic numbers 1 through 118 in the IUPAC periodic table. The requested extension consists of unconfirmed placeholders with atomic numbers 119 through 138. The extension is an indexing convention for hypothetical candidates, not a statement that these elements have been discovered or possess known chemistry. [IUPAC periodic table](#)

[0005] Four evidence labels are used locally: ESTABLISHED for an attributed source teaching; CALCULATED for a stated mathematical result or model calculation; PROPOSED for an unperformed preparation or test with specified assumptions; and UNSUPPORTED for a possibility without adequate preparation or property support. These labels do not decide patentability. A calculated fraction is not a measured composition, and a proposed recipe is not a performed experiment.

[0006] No new synthesis, measured yield, verified phase, measured property, or applicant-specific successful embodiment is reported in this draft. Published third-party examples retain their attribution. The requested breadth includes unsupported candidates. These candidates are expressly identified so that the scope is visible without representing them as established teachings.

3 Summary of the composition framework

[0007] A candidate material is described by a record $R = (B, S, x, I, U, C, P, M, F, E)$, where B is the composition basis; S is the constituent support set; x is the normalized composition vector; I is the isotope and nuclear-state description; U is the structural and spatial description; C is the preparation and conditioning record; P is the property scenario; M is the measurement record; F identifies failure boundaries; and E identifies evidence and sources. Missing fields are stated explicitly.

[0008] The framework includes every nonempty subset of the 118 recognized element labels and, separately, every nonempty subset of the 138-label vocabulary including the hypothetical extension. It includes every positive normalized continuous composition as a target description and every rational normalized composition as a separate exact count-compatible description. A finite decimal grid is a useful additional subset, not a replacement for either domain.

[0009] Each interval embodiment is the intersection of its component bounds with the normalization condition. Component endpoints cannot be selected independently if their sum would differ from one. Numerical enumeration provides a reproducible address for a candidate; it supplies neither missing chemistry nor a proof that later claims to that candidate lack novelty.

4 Composition bases and constituent accounting

[0010] On an atomic basis, $x_i = N_i / \sum(N_j)$, where N_i counts atoms of element i in the specified inventory. A mole-of-atoms basis has the same normalized proportions when the same atom inventory is used. A mole-of-molecules, formula-unit, repeat-unit, site-occupancy, or feed-component basis is different and must be expressly named. Denominators must identify the counted entity, sample region, and inclusion of impurities.

[0011] On a mass basis, $w_i = m_i / \sum(m_j)$. Conversion from atomic fractions uses $w_i = x_i M_i / \sum(x_j M_j)$, with M_i appropriate to the specified isotope distribution and units. The inverse uses $x_i = (w_i / M_i) / \sum(w_j / M_j)$. Standard atomic weights are not substitutes for unknown masses of hypothetical nuclei. Conversion is unavailable where the needed mass information is unavailable.

[0012] Volume fractions describe identified phases or constituents under specified temperature, pressure, density, and volume-assignment conventions. They are not automatically elemental atomic fractions. Mixing volumes need not be additive. A composite record therefore identifies constituent densities and whether volume fractions are measured, calculated from a justified additive model, or nominal feed values.

[0013] Feed proportions, retained product composition, and local measured composition are separate fields. A specimen may contain gradients, layers, inclusions, or surface enrichment. The support set identifies which constituents have positive fractions in the stated denominator. Zero entries are omitted from the canonical support set, while a zero concentration endpoint can describe a boundary shared with a smaller support set.

[0014] A closed composition specification includes all constituents in its denominator. An open specification can permit additional constituents, but must state how their inclusion affects normalization. A record describing selected constituents relative only to their subtotal is not a complete bulk-composition assay. Residual impurities are included as identified amounts, bounded unknown residuals, or an expressly unresolved term; they are not silently set to zero.

5 Exact constituent support and composition domains

[0015] Let $E = \{E_1, \dots, E_{118}\}$, with E_Z corresponding to the recognized element of atomic number Z . Let $H = \{E_{119}, \dots, E_{138}\}$ denote the hypothetical placeholders. For either vocabulary V , choose an ordered support $S = (E_{z1}, \dots, E_{zk})$, with strictly increasing atomic numbers. The ordering removes duplicate permutations; it does not restrict spatial ordering within a material.

[0016] For each integer k from 1 through the vocabulary size, the continuous target domain is $\Delta(S) = \{x : x_i > 0 \text{ for every member of } S, \sum(x_i) = 1\}$. Its closure permits $x_i = 0$, with the resulting candidate reassigned to its positive support. Every value is dimensionless; percent values equal $100 x_i$. For $k = 1$, the only normalized vector is (1).

[0017] The rational domain consists of all $x_i = n_i/q$, where q is any positive integer, each n_i is a positive integer, and $\sum(n_i) = q$. For k constituents, q must be at least k . A primitive representation has $\gcd(q, n_1, \dots, n_k) = 1$. This removes duplicate representations such as (1/2, 1/2) and (2/4, 2/4). The unrestricted denominator includes (1/3, 2/3), which is absent from a grid whose denominator is 10^{19} .

[0018] The requested decimal count grid has $D = 10^{19}$. Its vectors satisfy $x_i = n_i/D$, $n_i \geq 1$, and $\sum(n_i) = D$. The smallest positive grid fraction is $a = 1/10^{19} = 0.0000000000000000001$. In percent units it is 0.000000000000000001% . This is a chosen grid increment, not a universal physical concentration limit.

[0019] For k positive constituents on this grid, no single constituent can exceed $1 - (k-1)/D$. A two-component extreme is $(1/D, (D-1)/D)$. A 138-component extreme has 137 constituents each at $1/D$ and the remaining constituent at $(D-137)/D$. These vectors normalize exactly. They do not establish that a chemically homogeneous or stable 138-element material can be made.

[0020] The broad continuous and unrestricted rational domains remain available below the selected grid floor. If a particular embodiment imposes a positive floor a , then $ka \leq 1$ is necessary and sufficient for an unconstrained normalized vector to exist. For a fixed total atom inventory N , the least positive atomic count fraction is $1/N$. Since N depends on the sample, no sample-independent lower fraction follows from atom counting.

[0021] The SI Avogadro constant is exactly $6.02214076 \times 10^{23}$ per mole. An inventory of 10^{19} atoms is approximately 0.00001660539067 mole of atoms. One specified atom among that inventory gives a count fraction of 10^{-19} . The arithmetic does not establish controlled placement, detectability, representative sampling, or a reproducible bulk property. [BIPM definition of the mole](#)

[0022] The recognized vocabulary has $2^{118} - 1$ nonempty supports. Its supports of size at least two number $2^{118} - 1 - 118$. The extended vocabulary has $2^{138} - 1$ nonempty supports and $2^{138} - 1 - 138$ supports of size at least two. For a fixed selected support of size k on the D grid, the number of positive vectors is $\text{binomial}(D-1, k-1)$. These are counts of mathematical descriptions, not counts of enabled substances.

6 All endpoint ranges and complementary proportions

[0023] For each precision index $p = 0, 1, \dots, 19$, let $Q_p = 10^p$. Select integers l and u with $0 \leq l \leq u \leq Q_p$. The closed interval $I(p, l, u) = [l/Q_p, u/Q_p]$ defines every pair of decimal-grid endpoints at that precision, including exact-point intervals $l = u$. Percent endpoints are $[100 l/Q_p, 100 u/Q_p]$. Every elementary cell $[j/Q_p, (j+1)/Q_p]$ is included for $0 \leq j < Q_p$, as is every union represented by a contiguous pair of endpoints.

[0024] At $p = 2$ the elementary cells are $[0\%, 1\%]$, $[1\%, 2\%]$, ..., $[99\%, 100\%]$. For a binary composition, if x_A lies in $[L, U]$, $x_B = 1 - x_A$ lies in $[1-U, 1-L]$. Thus $[1\%, 2\%]$ pairs with $[98\%, 99\%]$, $[2\%, 3\%]$ with $[97\%, 98\%]$, and $[49\%, 50\%]$ with $[50\%, 51\%]$. Corresponding endpoints describe correlated compositions: (1%, 99%) and (2%, 98%), not (1%, 98%).

[0025] Closed cells share their endpoints. This intentional overlap avoids losing boundary values. Canonical indexing may assign a shared boundary to one cell for retrieval, but the physical composition is unchanged. Exact boundary points, every intermediate target value, each contiguous subrange with specified endpoints, and each complement are contemplated as numerical descriptions. This language does not assert that all such subranges are legally disclosed with sufficient specificity.

[0026] The requested interval 0.0000000000000001% through 0.0000000000000001% equals $[10^{-18}, 10^{-17}]$ as a fraction. On the D grid its numerators are 10 through 100 inclusive: 91 exact grid values and 90 elementary bins. The binary complement is $[(D-100)/D, (D-10)/D]$, or $[99.99999999999999\%, 99.99999999999999\%]$. Fraction and percent units must not be interchanged.

[0027] The 50/50 binary vector is exactly $(1/2, 1/2)$. The entire decimal hierarchy connects ordinary percent cells to the requested very small cells. Additional rational intervals $[r, s]$, for all rational $0 \leq r \leq s \leq 1$, and real endpoint target intervals may be specified outside the decimal hierarchy. Real targets, rational finite-count descriptions, decimal grid values, and measured tolerances remain distinct.

[0028] A decade refinement supplies optional addresses for very small positive values. For e from -19 through -1 and any nonnegative integer r , define intervals $[j \times 10^{(e-r)}, (j+1) \times 10^{(e-r)}]$ with $j = 10^r, \dots, 10^{(r+1)}-1$. These partition each selected decade at the specified refinement, with shared closed boundaries. Refinement beyond the D grid is symbolic or rational coverage, not an assertion of greater attainable manufacturing precision.

[0029] For a selected k -component support choose an interval $[L_i, U_i]$ for every component, with $0 \leq L_i \leq U_i \leq 1$, after intersecting any supplied bounds with $[0, 1]$. The candidate domain is $\{x : L_i \leq x_i \leq U_i, \sum(x_i) = 1\}$, intersected with the applicable positivity or grid conditions. A box with closed component bounds is feasible on the nonnegative continuous simplex exactly when each $L_i \leq U_i$ and $\sum(L_i) \leq 1 \leq \sum(U_i)$. This test is applied after intersecting with any selected positive floor and corresponding upper bound.

[0030] A witness can be constructed exactly: start with $x_i = L_i$; let residual $r = 1 - \sum(L_i)$; in support order add $\min(r, U_i - L_i)$ to each component and reduce r accordingly. The box is feasible if the residual reaches zero. With rational inputs the calculation uses integer fractions. Strict positivity without a fixed floor needs a separate check; a greedy witness containing zero does not prove that a positive witness is unavailable. With the ordinary feasibility conditions satisfied, require every $U_i > 0$. If $\sum(L_i) = 1$, feasibility with this support additionally requires every $L_i > 0$. If $\sum(L_i) < 1$, choose $\epsilon = \min(\min_i U_i, (1 - \sum(L_i))/(2k))$, replace each lower bound by $\max(L_i, \epsilon)$, and perform the residual allocation. ϵ is positive and the raised lower-bound sum remains below one, so the construction returns a positive witness. With rational inputs all operations remain exact. A particular vector having zero entries is reassigned to its smaller positive support.

[0031] For the D grid, set $b_i = \max(1, \text{ceiling}(D L_i))$ and $c_i = \min(D-k+1, \text{floor}(D U_i))$, and test $b_i \leq c_i$ and $\sum(b_i) \leq D \leq \sum(c_i)$. The same residual allocation uses integers. A box may be continuously feasible and contain no D-grid vector. For example the exact binary vector $(1/3, 2/3)$ is rationally feasible but is not on the D grid.

[0032] Independent selection of all endpoint ranges is permitted as an indexing operation, but infeasible combinations are retained only as rejected numerical requests, with the reason recorded. Example: three lower bounds of 40% cannot normalize. A valid ternary example has atomic intervals $A = [10\%, 20\%]$, $B = [30\%, 40\%]$, $C = [40\%, 60\%]$, with witness $(10\%, 30\%, 60\%)$. No product phase or preparation follows from that witness.

7 Structural and spatial descriptions

[0033] A material record identifies whether the composition describes a single phase, multiple phases, a molecular species, a solution, a dispersed mixture, a layered article, or an ensemble. Where the distinction is unknown, it is marked unknown. A numerical vector does not imply every constituent occupies a common lattice or remains uniformly distributed.

[0034] For a crystalline candidate, specify the phase identity, space group or structural model where known, lattice parameters with conditions, occupied sites, substitutions, vacancy rules, ordering, and charge-balance constraints. A proposed site assignment is labeled PROPOSED. Unknown structure is not filled by saying all crystal structures are present.

[0035] For an amorphous candidate, identify the preparation history, detectable crystalline fraction, relevant local-order evidence, glass transition or relaxation behavior where measured, and the test's resolution. For molecular candidates identify connectivity, stereochemistry, salt/solvate and hydration state, and purity. Distinct connectivity is separately indexed even if elemental fractions match.

[0036] For polymers specify repeat units, incorporated comonomer ratios, sequence distribution, tacticity, molecular-weight distribution, branching, end groups, crosslink topology, additives, and conditioning. Feed composition alone does not establish incorporated composition. For composites specify constituent identities, dimensions, orientation, interfaces, dispersion, layer order, void fraction, and the basis of each fraction.

[0037] Spatial variations can be represented by a composition field $x_i(r,t)$ with a defined coordinate region and sampling or averaging convention. A whole-specimen fraction is an integral over the appropriate count or mass density, not a simple average of unequal regions. Gradient, core-shell, layered, patterned, and porous descriptions require their dimensions and fabrication information. None is assumed to arise from every overall vector.

8 Preparation and confirmation records

[0038] Each preparation record states starting substances, grades, quantities, apparatus, order of operations, temperature, pressure, atmosphere, time, cooling or curing history, isolation, purification, post-treatment, scale, and relevant storage. Distinguish actual parameters from suggested values. A route class such as melting or deposition is insufficient as the preparation instruction for an unfamiliar composition.

[0039] Metals and alloys may be investigated through documented melting, powder consolidation, electrodeposition, or vapor deposition, followed by specified thermal and mechanical treatment. The selected route must address volatilization, immiscibility, reaction, contamination, competing phases, and phase stability for the actual constituents. This is a route-selection framework, not an assertion that one process works for every support.

[0040] Inorganic and ceramic candidates may use documented solid-state, solution, hydrothermal, solvothermal, deposition, or precursor-conversion routes. The actual precursors, stoichiometry, atmosphere, thermal schedule, isolation and phase-confirmation steps are necessary. Molecular candidates require a defined reaction or assembly and purification. Polymers require a defined polymerization, incorporated composition and forming history. Composites require constituent preparation, mixing or assembly, consolidation, and interface controls.

[0041] An unperformed recipe is a PROPOSED example stated prospectively. It reports no fabricated yield, measurement, or success. If a step depends on discovering an unavailable precursor, synthesizing a hypothetical element, stabilizing an unobserved nucleus, or establishing an unknown reaction, that dependency is an unresolved research problem rather than routine implementation.

[0042] Confirmation is selected to distinguish the target from realistic alternatives. Composition analysis alone may not identify a polymorph; diffraction alone may not detect a trace constituent; nominal instrument resolution is not a validated limit of detection in the actual matrix. Record calibration, reference materials, uncertainty, sampling, specimen history, and the method's detection and quantification limits. A nondetection is not proof of absence. [IUPAC limit of detection](#)

[0043] A tiny exact fraction is a mathematical target until preparation and measurement support a physical assertion. Experimental tolerance is specified separately, for example an interval or uncertainty with confidence convention. No instrument is claimed here to distinguish compositions separated by 10^{-19} fraction. At trace levels, isotope signatures, contamination control, adsorption, sampling statistics, and atom-count discreteness can dominate the interpretation.

[0044] An established teaching for comparison is TiO₂(B) nanosheets reported by Xiang and colleagues in 2013. Their nominal formula has Ti:O = 1:2; the B phase, nanosheet morphology, and exposed surface are additional features. Their paper reports a solvothermal preparation from titanium tetrachloride, water, and ethylene glycol, at 150 degrees C for six hours, with isolation and washing and diffraction/microscopy characterization. This is an attributed source summary; the complete methods remain necessary, and no applicant repetition is claimed. [Xiang et al., Scientific Reports 3, 1411](#)

[0045] The TiO₂ formula corresponds to rational atomic fractions (1/3,2/3). Its inclusion demonstrates why the unrestricted rational domain is needed and why a decimal grid alone is incomplete. It does not establish exact product purity, all titanium oxide polymorphs, every Ti/O ratio, or every process and use. Reproducing old literature is not represented as creating a newly available teaching.

9 Isotopes and nuclear states

[0046] An isotope-resolved constituent is identified by atomic number Z , mass number A , neutron count $N = A - Z$, and nuclear-state designation. For the elemental vocabulary, Z is 1 through 118 or an explicitly hypothetical 119 through 138. A is an integer at least Z . This parameterization includes candidate identifiers beyond observed nuclides; it does not assert that every (Z,A) nucleus is bound, observable, stable, or preparable.

[0047] For each element i , let $y_{i,a,s}$ be the conditional fraction of its atoms assigned to isotope a and nuclear state s , with $y \geq 0$ and $\sum(y_{i,a,s}) = 1$ over the stated isotope inventory. The whole-sample atomic fraction of that nuclide-state is $x_i y_{i,a,s}$. Natural, enriched, depleted, pure-isotope target, and mixed-isotope distributions can be described. Availability, lifetime, decay products, state preparation, and time since preparation are separate requirements.

[0048] Ground states, isomers, resonances, and evaluated excited levels are not collapsed into one identifier. NUBASE2020 and AME2020 provide dated evaluation resources for nuclear states and atomic masses. Their measured, estimated, and unobserved entries must retain those distinctions. The 2020 edition's experimental cutoff is not a statement that it contains every nuclide known in September 2026. [NUBASE2020](#), [AME2020 Part I](#), [AME2020 Part II](#)

[0049] An archival isotope record retains the source edition, nuclide identifier, state, existence-status evidence, property-estimation markers, units, uncertainties, half-life, decay modes, and retrieval date. A systematics marker on a mass is not by itself an existence-status classification. LiveChart API records likewise require their evaluation provenance; its guide identifies the underlying snapshots. No complete current nuclide export is represented as included in this draft. [IAEA LiveChart API guide](#)

[0050] A changing radioactive inventory is a function of time and preparation history. Its composition vector is identified at a specified time or over a stated averaging period. Daughter products enter the support when

included in the denominator. A short-lived nuclear species is not assumed to persist through an ordinary bulk preparation or measurement.

10 Subatomic and emergent particle systems

[0051] Established elementary-particle categories are distinguished from composite particles, nuclei, material excitations, and hypothetical particles. The Standard Model register includes six quark flavors, six lepton flavors, gauge bosons, and the Higgs boson, with antiparticle and charge-state conventions where applicable. A category listing does not teach arbitrary stable macroscopic mixtures of those particles. [CERN Standard Model](#)

[0052] A particle-system record identifies species and experimental status, quantum numbers, charge, mass where established, energy/momentum distribution, population or occupation convention, confinement or host, temperature or non-equilibrium state, lifetime, interaction model, preparation, and observation. Quarks within a nucleus, that nucleus, and the atom containing it are not independently added to an elemental count unless a physical construction and separate accounting basis are defined.

[0053] Composite particles such as protons, neutrons, mesons, and nuclei require their own identities and preparation or host descriptions. Photons and other field excitations require frequency or energy and occupation conditions rather than an unqualified chemical mass fraction. Antimatter-containing candidates require containment and annihilation considerations; no ordinary stable alloy is inferred from a particle list.

[0054] Phonons, magnons, excitons, polarons, and related quasiparticles are host-dependent excitations. Their record specifies the underlying material, excitation mechanism, field or carrier conditions, lifetime, and observable. They do not extend the periodic table and are not interchangeable elemental feedstocks. [IUPAC quasiparticle definition](#)

[0055] Hypothetical classes include gravitons, axion-like particles, sterile-neutrino candidates, dark photons, magnetic-monopole candidates, supersymmetric partners, and other model-defined entities. Each is an UNSUPPORTED candidate unless evidence and a preparation/observation route are supplied. A model identifier, parameters, conservation constraints, and search status must accompany it. No finite list can establish all future theoretical particle species or enable their material combinations.

[0056] PDG2026 tables and the versioned PDG API are source locators for particle identities and properties. They include experimental searches and limits as well as established particles. Inclusion in a table is not proof of discovery. The cited 2026 listings have a stated cutoff, and updates need their own date. [PDG 2026 tables](#), [PDG API](#)

11 Conditional properties and alternative responses

[0057] A property scenario identifies a baseline material and state, a controlled change, the expected response with mechanism and conditions, a credible alternative response with its mechanism, a discriminating test, and evidence status. The baseline and alternative are separate conditional embodiments or hypotheses. They are not asserted simultaneously for the same specimen under identical conditions.

[0058] Mechanical scenarios include stiffness, shear modulus, yield and ultimate strength, torsional response, fracture toughness, ductility, hardness, fatigue, creep, wear, and stress relaxation. These quantities have distinct definitions. A proposed strength increase cannot be inferred from stiffness alone, and specimen geometry can affect torque resistance. Tests specify direction, temperature, strain rate, environment, conditioning, and failure criterion.

[0059] Electrical scenarios include DC and AC conductivity, resistivity, carrier type and density, mobility, dielectric response, capacitance, breakdown, piezoresponse, thermoelectric response, and superconducting behavior where supported. Frequency, field, temperature, contacts, and geometry are specified. Negative differential resistance under driven conditions is not treated as negative passive DC conductivity. No assertion of superconductivity or a transition temperature is made for an unidentified composition.

[0060] Additional scenario fields include thermal conductivity and expansion; heat capacity and transitions; optical absorption, emission, refractive and nonlinear response; magnetic ordering, susceptibility and coercivity; corrosion, oxidation and chemical stability; catalytic selectivity and rate; diffusion, sorption, permeability and wetting; radiation response; and biological interaction where relevant. Each requires a defined observable, conditions, model or measurement, and local evidence label. A category name is not a predicted performance value.

[0061] A scenario may propose increased, decreased, approximately unchanged, threshold, nonmonotonic, hysteretic, anisotropic, transient, or sign-changing response, only with an explicit baseline and physically meaningful definition. An unchanged result means within stated resolution or tolerance, not mathematically identical under every test. These response categories enumerate research questions; they do not establish all future unexpected results.

[0062] ESTABLISHED mechanism example with PROPOSED extrapolation: in suitable precipitation-strengthened alloys, fine precipitates can impede dislocations; altered aging can coarsen precipitates and reduce hardness. For a new alloy record, specify its phases and aging path, then test hardness and microscopy rather than asserting every added element strengthens it. [NBS study of 2024 aluminum alloy processing](#)

[0063] ESTABLISHED mechanism example with PROPOSED extrapolation: a cooling path in a glass-forming system may suppress crystallization, while a different path or geometry permits nucleation and growth. A candidate record specifies the melt and measured thermal history and distinguishes glass from crystalline product using diffraction and calorimetry. This does not identify glass-forming behavior for every support. [Johnson et al., metallic-glass formation](#)

[0064] ESTABLISHED mechanism example with PROPOSED extrapolation: particular defects in TiO₂ can affect carrier separation and photocatalytic pathways, while trapping and recombination can lower useful activity. A new proposal must identify the defect location and material state and separately measure carrier dynamics and reaction rate. Increased absorption is not automatically increased photocatalysis. [Defects and photocatalysis in rutile TiO₂](#)

[0065] ESTABLISHED mechanism example with PROPOSED extrapolation: effective polymer crosslinks can increase stiffness, but nominal added crosslinker does not determine the number of active crosslinks or network architecture. A proposed opposing response is tested through conversion, swelling, and modulus at controlled temperature and frequency. [Crosslinking of fibrous hydrogels](#)

[0066] ESTABLISHED mechanism example with PROPOSED extrapolation: conductive fillers can form transport networks; dispersion, contact barriers, and competing network mechanisms can change conductivity versus loading. The same named constituents do not fix the percolation threshold. A proposal records morphology and contact conditions alongside electrical measurements. [Competing electrical-percolation behavior](#)

[0067] ESTABLISHED mechanism example with PROPOSED extrapolation: aggregation can increase emission for suitable luminogens by changing nonradiative pathways, while other packing and excited-state interactions can quench it. A new candidate needs molecular identity, packing evidence, quantum yield, lifetime, and conditions. The alternatives do not apply indiscriminately to every compound. [Tetrathienylethene emission study](#)

[0068] For an otherwise identical material, an unrecognized property may already be inherent, but necessity must be established for the relevant conditions. Stating a property and its opposite as possible does not establish inherency. A later process or use with additional steps is separately compared. The property-scenario framework is not a promise to prevent patents on every later finding. [USPTO inherency guidance](#)

12 Worked numerical embodiments

Binary percent cells

[0069] CALCULATED example NUM-001 selects an arbitrary ordered pair of distinct constituent labels A and B, on an atomic basis. For every $j = 0, \dots, 99$, x_A is in $[j/100, (j+1)/100]$ and $x_B = 1 - x_A$. Every composition satisfying $0 < x_A < 1$ has two-element support, including shared internal cell endpoints. Only $x_A = 0$ or $x_A = 1$ is unary. No particular chemical preparation is inferred. The pair can be replaced by any ordered distinct pair from the specified vocabulary, with hypothetical labels retaining their status.

Fine trace range

[0070] CALCULATED example NUM-002 selects $x_A = 10/D, 11/D, \dots, 100/D$ and $x_B = 1 - x_A$, with $D = 10^{19}$. For instance $x_A = 10/D = 10^{-18}$ fraction = $10^{-16}\%$, and $x_B = (D-10)/D$. Exact fractions are stored as integer numerator and denominator strings, avoiding floating-point rounding of the complement. Every contiguous subrange $[l/D, u/D]$ with $10 \leq l \leq u \leq 100$ is separately addressable. Preparation and detectability remain UNSUPPORTED.

Rational composition outside the decimal grid

[0071] CALCULATED example NUM-003 uses $(1/3, 2/3)$. No integer n satisfies $n/10^{19} = 1/3$ because 3 does not divide 10^{19} . The example is explicitly included through unrestricted rational denominators. A molecular stoichiometry, phase, and product purity require additional descriptions; the vector alone is not the $\text{TiO}_2(\text{B})$ literature product.

Multiple components and rejected intervals

[0072] CALCULATED example NUM-004 uses the feasible ternary box in paragraph [0032]. NUM-005 rejects three lower bounds of 40% because their sum is 120%. NUM-006 selects any k from 2 through 138, assigns $1/D$ to $k-1$ constituents and $(D-k+1)/D$ to the last. For $k = 138$ the small constituents total $137/D$. An equal-fraction candidate uses $x_i = 1/k$ in the rational domain; it lies on the D grid only when k divides D .

13 Reproducible enumeration instructions

[0073] To enumerate supports, fix the vocabulary edition, choose k , and list every increasing k -tuple of atomic numbers. Attach each label's recognized or hypothetical status. To enumerate a fixed rational denominator q , list every positive integer k -tuple summing to q . Use the last numerator as q minus the sum of the preceding numerators and reject values below one. This procedure terminates for each finite q and k .

[0074] To enumerate the unrestricted rational domain, repeat for $q = k, k+1, \dots$ and retain only primitive representations when deduplication is required. This is a countably infinite sequence, not a claim that a finite application prints every tuple. To enumerate decimal intervals, iterate p , l , and u over the finite limits in paragraph [0023], form component boxes, apply normalization feasibility, and construct a witness. The real continuous domain is defined symbolically; it cannot be exhaustively listed by a finite digital enumeration.

[0075] Every generated record includes vocabulary edition, support tuple, composition basis, exact endpoints, normalization rule, witness or infeasibility reason, and evidence status. A record identifier encodes these

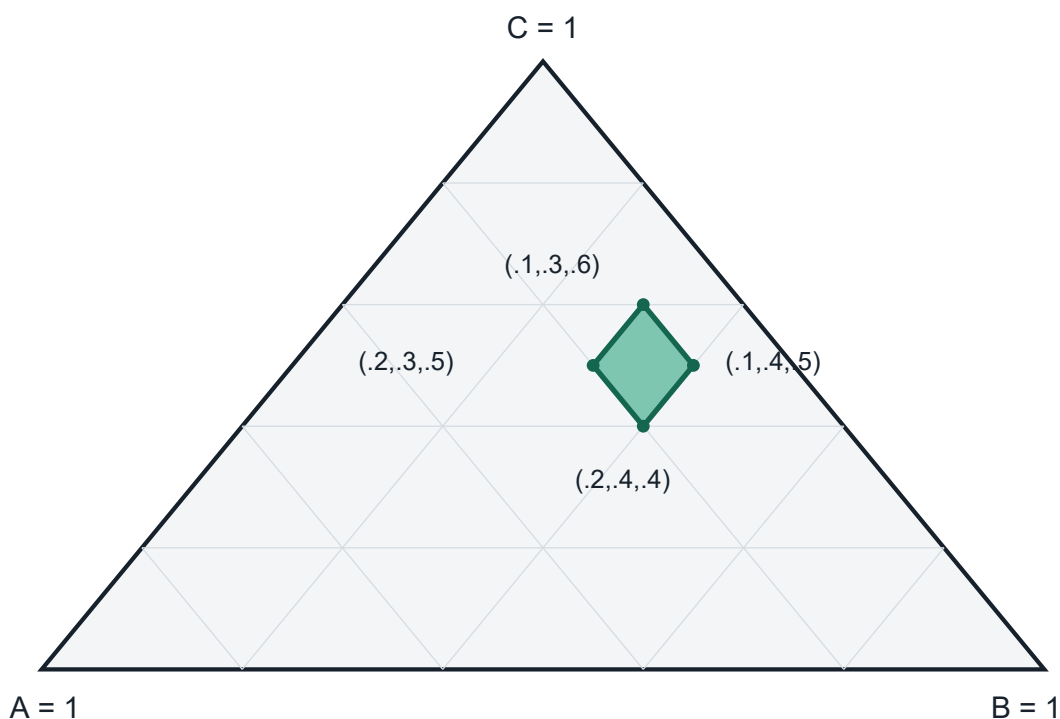
inputs, not an assertion of a synthesized material. Material identity, structure, preparation, and property fields are added independently and must not default to successful or established.

[0076] A reproducibility package accompanies this draft for exact arithmetic and verification. Its external files are aids for review, not automatically part of a filed or published application. Essential definitions and worked examples are therefore stated in this description. A final filing must carry all relied-on technical content in accepted application formats, with the filed and published versions checked.

14 Drawings

[0077] Figure 1 illustrates normalization for three constituents. The triangle represents the nonnegative simplex; its vertices are unary boundaries. The highlighted feasible interval intersection represents A in [10%,20%], B in [30%,40%], and C in [40%,60%]. It is a numerical diagram, not a measured phase diagram.

Figure 1. Normalized three-component composition

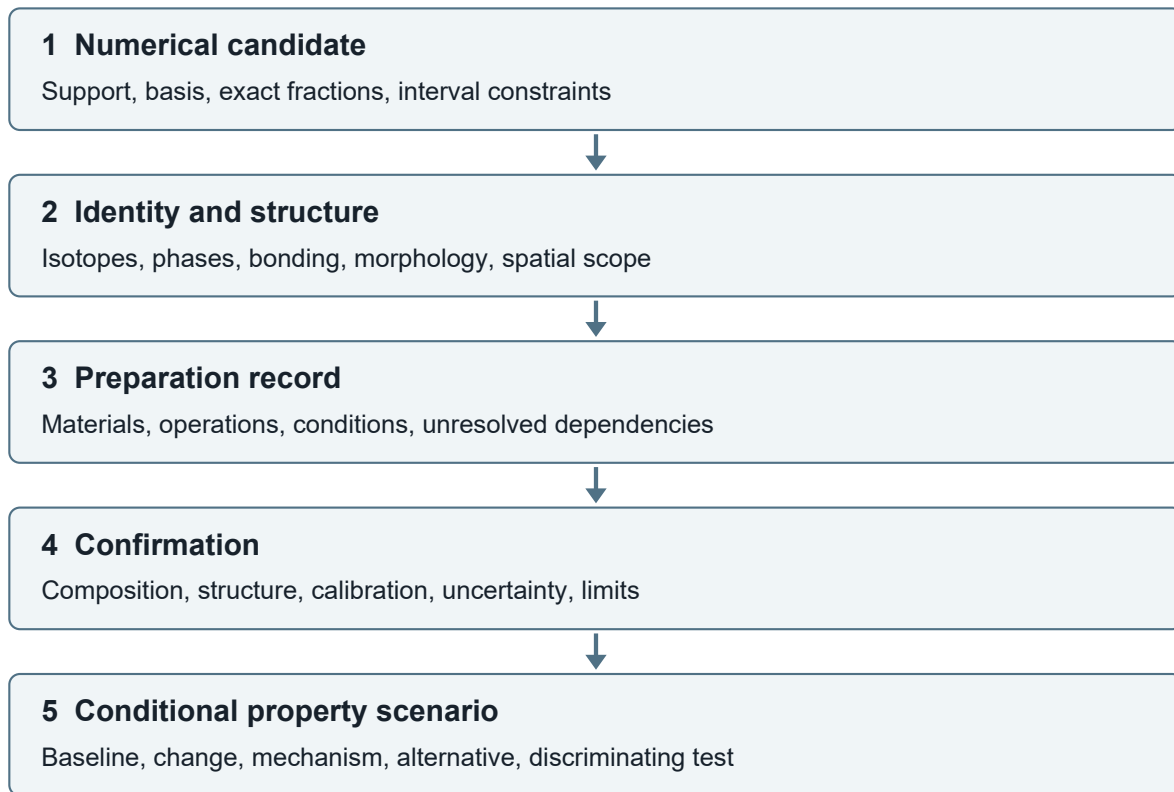


Highlighted: A in [0.1,0.2], B in [0.3,0.4], C in [0.4,0.6].

Every point obeys $A + B + C = 1$. Numerical region only; not a phase diagram.

[0078] Figure 2 shows the material-record sequence from a numerical candidate to structure, preparation, confirmation, and a conditional property scenario. An unresolved field remains unresolved. The drawing does not assert that this workflow supplies a universal preparation route.

Figure 2. Candidate-material record sequence



At every step: preserve evidence status and leave missing information unresolved.

A completed numerical record is not proof of an available physical material.

15 Industrial applicability and unresolved technical scope

[0079] Supported individual materials may be applicable to structural components, electronic and optical devices, thermal management, chemical processing, coatings, storage, or other uses established for the individual record. This draft does not establish industrial applicability of every hypothetical element, nuclide, particle system, or composition. An actual material-use record must state how the identified product is made and used.

[0080] Unresolved technical scope includes preparation of most indexed compositions; crystal and molecular identity; phase stability and coexistence; isotope availability and lifetime; precision and impurity control; particle confinement; property magnitudes and conditions; and the applicant's actual technical contribution. Numeric refinement cannot resolve these gaps. The candidate claims below express the requested scope for review and are not certified as novel, enabled, unified, or allowable.

16 Draft abstract

A framework describes candidate materials through elemental support sets, normalized composition vectors, interval constraints, isotope distributions, structures, preparation records, and conditional property scenarios. The vocabulary separates 118 recognized elements from hypothetical atomic-number placeholders through 138. Composition descriptions include positive continuous targets, unrestricted rational proportions, a decimal count grid with denominator 10^{19} , and correlated component ranges intersected with normalization. Exact endpoints, binary complements, and feasibility witnesses support reproducible indexing. Nuclear states and particle systems use separate identity and accounting conventions. Material records associate numerical candidates with preparation, characterization, evidence status, and unresolved conditions. The framework distinguishes calculated descriptions and unperformed proposals from established material teachings.

17 Candidate claims

The following complete claim annex expresses requested coverage for review. Hypothetical constituents and broad material classes lack demonstrated full-scope preparation support. Adding separate count or interval claims does not cure that deficiency.

Novelty has not been assessed for any of claims 1-199. Claims 1-185 are physical-material review candidates; claims 186-193 and 197-199 concern record operations, and claims 194-196 concern separate particle-system records. No claim represents an assertion that its full physical scope is enabled.

Claim 1. A material comprising a closed elemental inventory of k distinct elemental constituents selected from atomic numbers 1 through 138, wherein k is an integer from 2 through 138, each selected constituent has a positive normalized fraction x_i on an expressly identified atomic-fraction or mass-fraction basis, the fractions of the selected constituents sum to one, and constituents outside the selected inventory have zero fraction on that basis.

Claim 2. The material of claim 1, wherein k is exactly 2.

Claim 3. The material of claim 1, wherein k is exactly 3.

Claim 4. The material of claim 1, wherein k is exactly 4.

Claim 5. The material of claim 1, wherein k is exactly 5.

Claim 6. The material of claim 1, wherein k is exactly 6.

Claim 7. The material of claim 1, wherein k is exactly 7.

Claim 8. The material of claim 1, wherein k is exactly 8.

Claim 9. The material of claim 1, wherein k is exactly 9.

Claim 10. The material of claim 1, wherein k is exactly 10.

Claim 11. The material of claim 1, wherein k is exactly 11.

Claim 12. The material of claim 1, wherein k is exactly 12.

Claim 13. The material of claim 1, wherein k is exactly 13.

Claim 14. The material of claim 1, wherein k is exactly 14.

Claim 15. The material of claim 1, wherein k is exactly 15.

Claim 16. The material of claim 1, wherein k is exactly 16.

Claim 17. The material of claim 1, wherein k is exactly 17.

Claim 18. The material of claim 1, wherein k is exactly 18.

Claim 19. The material of claim 1, wherein k is exactly 19.

Claim 20. The material of claim 1, wherein k is exactly 20.

Claim 21. The material of claim 1, wherein k is exactly 21.

Claim 22. The material of claim 1, wherein k is exactly 22.

Claim 23. The material of claim 1, wherein k is exactly 23.

Claim 24. The material of claim 1, wherein k is exactly 24.

Claim 25. The material of claim 1, wherein k is exactly 25.

Claim 26. The material of claim 1, wherein k is exactly 26.

Claim 27. The material of claim 1, wherein k is exactly 27.

Claim 28. The material of claim 1, wherein k is exactly 28.

Claim 29. The material of claim 1, wherein k is exactly 29.

Claim 30. The material of claim 1, wherein k is exactly 30.

Claim 31. The material of claim 1, wherein k is exactly 31.

Claim 32. The material of claim 1, wherein k is exactly 32.

Claim 33. The material of claim 1, wherein k is exactly 33.

Claim 34. The material of claim 1, wherein k is exactly 34.

Claim 35. The material of claim 1, wherein k is exactly 35.

Claim 36. The material of claim 1, wherein k is exactly 36.

Claim 37. The material of claim 1, wherein k is exactly 37.

Claim 38. The material of claim 1, wherein k is exactly 38.

Claim 39. The material of claim 1, wherein k is exactly 39.

Claim 40. The material of claim 1, wherein k is exactly 40.

Claim 41. The material of claim 1, wherein k is exactly 41.

Claim 42. The material of claim 1, wherein k is exactly 42.

Claim 43. The material of claim 1, wherein k is exactly 43.

Claim 44. The material of claim 1, wherein k is exactly 44.

Claim 45. The material of claim 1, wherein k is exactly 45.

Claim 46. The material of claim 1, wherein k is exactly 46.

Claim 47. The material of claim 1, wherein k is exactly 47.

Claim 48. The material of claim 1, wherein k is exactly 48.

Claim 49. The material of claim 1, wherein k is exactly 49.

Claim 50. The material of claim 1, wherein k is exactly 50.

Claim 51. The material of claim 1, wherein k is exactly 51.

Claim 52. The material of claim 1, wherein k is exactly 52.

Claim 53. The material of claim 1, wherein k is exactly 53.

Claim 54. The material of claim 1, wherein k is exactly 54.

Claim 55. The material of claim 1, wherein k is exactly 55.

Claim 56. The material of claim 1, wherein k is exactly 56.

Claim 57. The material of claim 1, wherein k is exactly 57.

Claim 58. The material of claim 1, wherein k is exactly 58.

Claim 59. The material of claim 1, wherein k is exactly 59.

Claim 60. The material of claim 1, wherein k is exactly 60.

Claim 61. The material of claim 1, wherein k is exactly 61.

Claim 62. The material of claim 1, wherein k is exactly 62.

Claim 63. The material of claim 1, wherein k is exactly 63.

Claim 64. The material of claim 1, wherein k is exactly 64.

Claim 65. The material of claim 1, wherein k is exactly 65.

Claim 66. The material of claim 1, wherein k is exactly 66.

Claim 67. The material of claim 1, wherein k is exactly 67.

Claim 68. The material of claim 1, wherein k is exactly 68.

Claim 69. The material of claim 1, wherein k is exactly 69.

Claim 70. The material of claim 1, wherein k is exactly 70.

Claim 71. The material of claim 1, wherein k is exactly 71.

Claim 72. The material of claim 1, wherein k is exactly 72.

Claim 73. The material of claim 1, wherein k is exactly 73.

Claim 74. The material of claim 1, wherein k is exactly 74.

Claim 75. The material of claim 1, wherein k is exactly 75.

Claim 76. The material of claim 1, wherein k is exactly 76.

Claim 77. The material of claim 1, wherein k is exactly 77.

Claim 78. The material of claim 1, wherein k is exactly 78.

Claim 79. The material of claim 1, wherein k is exactly 79.

Claim 80. The material of claim 1, wherein k is exactly 80.

Claim 81. The material of claim 1, wherein k is exactly 81.

Claim 82. The material of claim 1, wherein k is exactly 82.

Claim 83. The material of claim 1, wherein k is exactly 83.

Claim 84. The material of claim 1, wherein k is exactly 84.

Claim 85. The material of claim 1, wherein k is exactly 85.

Claim 86. The material of claim 1, wherein k is exactly 86.

Claim 87. The material of claim 1, wherein k is exactly 87.

Claim 88. The material of claim 1, wherein k is exactly 88.

Claim 89. The material of claim 1, wherein k is exactly 89.

Claim 90. The material of claim 1, wherein k is exactly 90.

Claim 91. The material of claim 1, wherein k is exactly 91.

Claim 92. The material of claim 1, wherein k is exactly 92.

Claim 93. The material of claim 1, wherein k is exactly 93.

Claim 94. The material of claim 1, wherein k is exactly 94.

Claim 95. The material of claim 1, wherein k is exactly 95.

Claim 96. The material of claim 1, wherein k is exactly 96.

Claim 97. The material of claim 1, wherein k is exactly 97.

Claim 98. The material of claim 1, wherein k is exactly 98.

Claim 99. The material of claim 1, wherein k is exactly 99.

Claim 100. The material of claim 1, wherein k is exactly 100.

Claim 101. The material of claim 1, wherein k is exactly 101.

Claim 102. The material of claim 1, wherein k is exactly 102.

Claim 103. The material of claim 1, wherein k is exactly 103.

Claim 104. The material of claim 1, wherein k is exactly 104.

Claim 105. The material of claim 1, wherein k is exactly 105.

Claim 106. The material of claim 1, wherein k is exactly 106.

Claim 107. The material of claim 1, wherein k is exactly 107.

Claim 108. The material of claim 1, wherein k is exactly 108.

Claim 109. The material of claim 1, wherein k is exactly 109.

Claim 110. The material of claim 1, wherein k is exactly 110.

Claim 111. The material of claim 1, wherein k is exactly 111.

Claim 112. The material of claim 1, wherein k is exactly 112.

Claim 113. The material of claim 1, wherein k is exactly 113.

Claim 114. The material of claim 1, wherein k is exactly 114.

Claim 115. The material of claim 1, wherein k is exactly 115.

Claim 116. The material of claim 1, wherein k is exactly 116.

Claim 117. The material of claim 1, wherein k is exactly 117.

Claim 118. The material of claim 1, wherein k is exactly 118.

Claim 119. The material of claim 1, wherein k is exactly 119.

Claim 120. The material of claim 1, wherein k is exactly 120.

Claim 121. The material of claim 1, wherein k is exactly 121.

Claim 122. The material of claim 1, wherein k is exactly 122.

Claim 123. The material of claim 1, wherein k is exactly 123.

- Claim 124.** The material of claim 1, wherein k is exactly 124.
- Claim 125.** The material of claim 1, wherein k is exactly 125.
- Claim 126.** The material of claim 1, wherein k is exactly 126.
- Claim 127.** The material of claim 1, wherein k is exactly 127.
- Claim 128.** The material of claim 1, wherein k is exactly 128.
- Claim 129.** The material of claim 1, wherein k is exactly 129.
- Claim 130.** The material of claim 1, wherein k is exactly 130.
- Claim 131.** The material of claim 1, wherein k is exactly 131.
- Claim 132.** The material of claim 1, wherein k is exactly 132.
- Claim 133.** The material of claim 1, wherein k is exactly 133.
- Claim 134.** The material of claim 1, wherein k is exactly 134.
- Claim 135.** The material of claim 1, wherein k is exactly 135.
- Claim 136.** The material of claim 1, wherein k is exactly 136.
- Claim 137.** The material of claim 1, wherein k is exactly 137.
- Claim 138.** The material of claim 1, wherein k is exactly 138.
- Claim 139.** A single-element material having a closed elemental inventory consisting of one constituent selected from atomic numbers 1 through 138, wherein the constituent has normalized atomic fraction one and every other elemental constituent has fraction zero.
- Claim 140.** The material of claim 1, wherein every selected constituent has atomic number from 1 through 118.
- Claim 141.** The material of claim 1, wherein at least one selected constituent is selected from atomic numbers 119 through 138.
- Claim 142.** The material of claim 1, wherein k is 118 and the inventory includes each atomic number from 1 through 118.
- Claim 143.** The material of claim 1, wherein k is 138 and the inventory includes each atomic number from 1 through 138.
- Claim 144.** The material of claim 1, wherein each x_i is the number of atoms of constituent i divided by the total number of atoms in the specified inventory.
- Claim 145.** The material of claim 1, wherein the proportions are specified as mole-of-atoms fractions using the same atom inventory as an atomic-fraction description.
- Claim 146.** The material of claim 1, wherein each x_i is the mass of constituent i divided by the total mass of the specified inventory.
- Claim 147.** The material of claim 1, wherein an isotope-qualified mass M_i is specified for every constituent and the atomic and mass fractions are related by $w_i = x_i M_i / \sum(x_j M_j)$.
- Claim 148.** The material of claim 1, wherein $x_i = n_i/q$ for a positive integer q and positive integers n_i whose sum is q, with q unrestricted except that q is at least k.
- Claim 149.** The material of claim 148, wherein $\gcd(q, n_1, \dots, n_k) = 1$.

Claim 150. The material of claim 1, wherein $x_i = n_i/D$ with $D = 10^{19}$, each n_i is a positive integer, and $\sum(n_i) = D$.

Claim 151. The material of claim 150, wherein $k-1$ constituents each have fraction $1/D$ and the remaining constituent has fraction $(D-k+1)/D$.

Claim 152. The material of claim 1, wherein each fraction satisfies a selected closed interval $[L_i, U_i]$, with $0 \leq L_i \leq U_i \leq 1$, and the joint composition satisfies $\sum(x_i) = 1$ and $x_i > 0$.

Claim 153. The material of claim 152, wherein each interval is selected by integers p, l, u with $0 \leq p \leq 19$ and $0 \leq l \leq u \leq 10^p$, its endpoints being $l/10^p$ and $u/10^p$.

Claim 154. The material of claim 153, wherein at least one interval is an elementary cell $[j/10^p, (j+1)/10^p]$ for an integer j with $0 \leq j < 10^p$.

Claim 155. The material of claim 1, wherein $k = 2$ and fractions x_A and x_B obey x_A in $[L, U]$ and $x_B = 1 - x_A$ in $[1 - U, 1 - L]$, with $0 < x_A < 1$.

Claim 156. The material of claim 155, wherein $10^{-18} \leq x_A \leq 10^{-17}$, corresponding to 10⁻¹⁶% through 10⁻¹⁵%, inclusive.

Claim 157. The material of claim 156, wherein x_A is exactly 10^{-18} and x_B is exactly $1 - 10^{-18}$.

Claim 158. The material of claim 156, wherein x_A is exactly 10^{-17} and x_B is exactly $1 - 10^{-17}$.

Claim 159. The material of claim 156, wherein $x_A = n/10^{19}$ for an integer n from 10 through 100 and $x_B = 1 - x_A$.

Claim 160. The material of claim 159, wherein x_A lies in a selected subrange $[l/10^{19}, u/10^{19}]$ for integers $10 \leq l \leq u \leq 100$.

Claim 161. The material of claim 1, wherein $k = 2$ and each constituent has fraction $1/2$, corresponding to 50% each.

Claim 162. The material of claim 1, wherein $k = 2$ and the constituent fractions are exactly $1/3$ and $2/3$.

Claim 163. The material of claim 1, wherein every constituent has fraction $1/k$.

Claim 164. The material of claim 1, wherein every constituent has fraction at least 10^{-19} .

Claim 165. The material of claim 1, wherein at least one constituent has positive fraction less than 10^{-19} .

Claim 166. The material of claim 152, wherein at least one interval is $[j \text{ times } 10^{(e-r)}, (j+1) \text{ times } 10^{(e-r)}]$, with integer e from -19 through -1, nonnegative integer r , and integer j from 10^r through $10^{(r+1)} - 1$.

Claim 167. The material of claim 152, wherein at least one interval has arbitrary rational endpoints r and s satisfying $0 \leq r \leq s \leq 1$.

Claim 168. The material of claim 152, wherein the endpoints are specified as real target values and the selected fractions satisfy their interval constraints and exact normalization.

Claim 169. The material of claim 1, wherein each selected element has a conditional isotope-state distribution $y_{i,a,s}$ with nonnegative entries summing to one and whole-inventory nuclide-state fractions $x_i y_{i,a,s}$.

Claim 170. The material of claim 169, wherein at least one constituent is specified by a single mass number and nuclear state with conditional fraction one.

Claim 171. The material of claim 169, wherein at least one constituent has two or more specified isotope-state entries of positive conditional fraction.

Claim 172. The material of claim 169, wherein each specified nuclear state has a source-qualified identifier that distinguishes a ground state from any specified isomer or excited state.

Claim 173. The material of claim 169, wherein a radioactive inventory is identified at a specified time and includes daughter constituents that fall within its stated denominator.

Claim 174. The material of claim 1, wherein a crystalline constituent phase has a specified structural model, site occupations, substitution rules and charge-balance constraints.

Claim 175. The material of claim 1, wherein an amorphous constituent phase is identified by a stated preparation history and a specified characterization convention for detectable crystalline fraction.

Claim 176. The material of claim 1, wherein molecular identity includes specified connectivity, stereochemistry, salt or solvate state, and hydration state where present.

Claim 177. The material of claim 1, wherein the material is a polymer specified by repeat units, incorporated ratios, sequence distribution, molecular-weight distribution and network architecture.

Claim 178. The material of claim 1, wherein the material is a composite specified by constituent identities, dimensions, orientations, interfaces and dispersion.

Claim 179. The material of claim 1, wherein the material has a specified layer order and layer dimensions, with local composition distinguished from whole-specimen composition.

Claim 180. The material of claim 1, wherein the material has a specified spatial composition field $x_i(r,t)$ and a density-weighted inventory convention for deriving the whole-specimen fractions.

Claim 181. The material of claim 1, wherein the stated elemental inventory describes a specified single phase.

Claim 182. The material of claim 1, wherein the stated elemental inventory describes a specified multiphase assemblage and each phase is separately identified.

Claim 183. The material of claim 178, wherein constituent volume fractions and void fraction are specified separately from elemental atomic fractions under stated temperature, pressure, density and volume-assignment conventions.

Claim 184. The material of claim 1, wherein a nominal target fraction is accompanied by a separately specified measurement uncertainty or tolerance and the associated sampling convention.

Claim 185. The material of claim 1, wherein the product composition is specified separately from its feed proportions and from local measured composition.

Claim 186. A method of constructing a candidate-material record, comprising selecting a constituent support and composition basis; assigning positive normalized target fractions whose sum is one; recording structural, preparation, measurement and property fields; and recording an evidence status for each field without treating an unresolved field as a successful preparation.

Claim 187. The method of claim 186, comprising enumerating every increasing k-tuple of selected atomic numbers for a specified k from 1 through 138 and attaching a recognized or hypothetical status to each label.

Claim 188. The method of claim 186, comprising enumerating positive integer tuples with sum q for successive unrestricted positive denominators q, and retaining primitive common-denominator representations.

Claim 189. The method of claim 186, comprising enumerating positive integer tuples with sum $D = 10^{19}$ and storing their normalized fractions as exact integer numerator and denominator strings.

Claim 190. The method of claim 186, comprising selecting every endpoint pair l, u with $0 \leq l \leq u \leq 10^p$ for each p from 0 through 19, forming correlated component boxes, and rejecting boxes that fail the applicable normalization and positivity tests.

Claim 191. The method of claim 186, comprising testing $0 \leq L_i \leq U_i \leq 1$ and $\sum(L_i) \leq 1 \leq \sum(U_i)$, then constructing a nonnegative witness by starting at the lower bounds and allocating residual mass without exceeding any upper bound.

Claim 192. The method of claim 191, further comprising checking every $U_i > 0$ for a strictly positive support; when $\sum(L_i) = 1$ requiring every $L_i > 0$; and when $\sum(L_i) < 1$ raising the lower bounds to $\max(L_i, \epsilon)$, where $\epsilon = \min(\min_i U_i, (1 - \sum(L_i))/(2k))$, before residual allocation.

Claim 193. The method of claim 186, comprising calculating $b_i = \max(1, \text{ceiling}(D L_i))$ and $c_i = \min(D - k + 1, \text{floor}(D U_i))$ for $D = 10^{19}$, accepting a grid box only when $b_i \leq c_i$ and $\sum(b_i) \leq D \leq \sum(c_i)$, and constructing an integer witness within those bounds.

Claim 194. A method of constructing a particle-system record, comprising identifying a species or model-qualified candidate, experimental status, quantum numbers, population or occupation convention, energy conditions, host or confinement, lifetime information, preparation and observation fields, while specifying a construction level that avoids counting a particle and its containing composite as independent copies of the same inventory.

Claim 195. The method of claim 194, wherein the record concerns a host-dependent excitation and specifies the host material, excitation mechanism, conditions, lifetime or linewidth, and observable.

Claim 196. The method of claim 194, wherein an unconfirmed entity is marked hypothetical, with a model identifier, parameter assumptions, conservation constraints and experimental-search status, without assigning unsupported measured properties.

Claim 197. The method of claim 186, comprising recording a property scenario with a baseline material and state, controlled change, proposed response, mechanism, conditions, discriminating test and evidence status.

Claim 198. The method of claim 197, comprising recording a credible alternative response and its mechanism as a separate conditional hypothesis, rather than asserting opposite responses for the same specimen under identical conditions.

Claim 199. The method of claim 186, comprising labeling attributed source teachings ESTABLISHED, stated mathematical or model results CALCULATED, unperformed tests or preparations PROPOSED, and possibilities without adequate support UNSUPPORTED, and retaining source and condition information with the labels.

18 Element and particle register

The following 138 rows are self-contained identity entries. Recognized rows follow the [IUPAC table dated 4 May 2022](#), checked 29 September 2026. E119-E138 are local hypothetical placeholders, not accepted symbols. No row supplies a making method or atomic mass.

Z	Symbol / label	Name / local description	Status
1	H	hydrogen	Recognized
2	He	helium	Recognized
3	Li	lithium	Recognized
4	Be	beryllium	Recognized
5	B	boron	Recognized
6	C	carbon	Recognized
7	N	nitrogen	Recognized
8	O	oxygen	Recognized
9	F	fluorine	Recognized
10	Ne	neon	Recognized
11	Na	sodium	Recognized
12	Mg	magnesium	Recognized
13	Al	aluminium	Recognized
14	Si	silicon	Recognized
15	P	phosphorus	Recognized
16	S	sulfur	Recognized
17	Cl	chlorine	Recognized
18	Ar	argon	Recognized
19	K	potassium	Recognized
20	Ca	calcium	Recognized
21	Sc	scandium	Recognized
22	Ti	titanium	Recognized
23	V	vanadium	Recognized
24	Cr	chromium	Recognized
25	Mn	manganese	Recognized
26	Fe	iron	Recognized

Z	Symbol / label	Name / local description	Status
27	Co	cobalt	Recognized
28	Ni	nickel	Recognized
29	Cu	copper	Recognized
30	Zn	zinc	Recognized
31	Ga	gallium	Recognized
32	Ge	germanium	Recognized
33	As	arsenic	Recognized
34	Se	selenium	Recognized
35	Br	bromine	Recognized
36	Kr	krypton	Recognized
37	Rb	rubidium	Recognized
38	Sr	strontium	Recognized
39	Y	yttrium	Recognized
40	Zr	zirconium	Recognized
41	Nb	niobium	Recognized
42	Mo	molybdenum	Recognized
43	Tc	technetium	Recognized
44	Ru	ruthenium	Recognized
45	Rh	rhodium	Recognized
46	Pd	palladium	Recognized
47	Ag	silver	Recognized
48	Cd	cadmium	Recognized
49	In	indium	Recognized
50	Sn	tin	Recognized
51	Sb	antimony	Recognized
52	Te	tellurium	Recognized
53	I	iodine	Recognized
54	Xe	xenon	Recognized
55	Cs	caesium	Recognized

Z	Symbol / label	Name / local description	Status
56	Ba	barium	Recognized
57	La	lanthanum	Recognized
58	Ce	cerium	Recognized
59	Pr	praseodymium	Recognized
60	Nd	neodymium	Recognized
61	Pm	promethium	Recognized
62	Sm	samarium	Recognized
63	Eu	europium	Recognized
64	Gd	gadolinium	Recognized
65	Tb	terbium	Recognized
66	Dy	dysprosium	Recognized
67	Ho	holmium	Recognized
68	Er	erbium	Recognized
69	Tm	thulium	Recognized
70	Yb	ytterbium	Recognized
71	Lu	lutetium	Recognized
72	Hf	hafnium	Recognized
73	Ta	tantalum	Recognized
74	W	tungsten	Recognized
75	Re	rhenium	Recognized
76	Os	osmium	Recognized
77	Ir	iridium	Recognized
78	Pt	platinum	Recognized
79	Au	gold	Recognized
80	Hg	mercury	Recognized
81	Tl	thallium	Recognized
82	Pb	lead	Recognized
83	Bi	bismuth	Recognized
84	Po	polonium	Recognized

Z	Symbol / label	Name / local description	Status
85	At	astatine	Recognized
86	Rn	radon	Recognized
87	Fr	francium	Recognized
88	Ra	radium	Recognized
89	Ac	actinium	Recognized
90	Th	thorium	Recognized
91	Pa	protactinium	Recognized
92	U	uranium	Recognized
93	Np	neptunium	Recognized
94	Pu	plutonium	Recognized
95	Am	americium	Recognized
96	Cm	curium	Recognized
97	Bk	berkelium	Recognized
98	Cf	californium	Recognized
99	Es	einsteinium	Recognized
100	Fm	fermium	Recognized
101	Md	mendelevium	Recognized
102	No	nobelium	Recognized
103	Lr	lawrencium	Recognized
104	Rf	rutherfordium	Recognized
105	Db	dubnium	Recognized
106	Sg	seaborgium	Recognized
107	Bh	bohrium	Recognized
108	Hs	hassium	Recognized
109	Mt	meitnerium	Recognized
110	Ds	darmstadtium	Recognized
111	Rg	roentgenium	Recognized
112	Cn	copernicium	Recognized
113	Nh	nihonium	Recognized

Z	Symbol / label	Name / local description	Status
114	Fl	flerovium	Recognized
115	Mc	moscovium	Recognized
116	Lv	livermorium	Recognized
117	Ts	tennessine	Recognized
118	Og	oganesson	Recognized
119	E119	hypothetical element 119	Hypothetical placeholder
120	E120	hypothetical element 120	Hypothetical placeholder
121	E121	hypothetical element 121	Hypothetical placeholder
122	E122	hypothetical element 122	Hypothetical placeholder
123	E123	hypothetical element 123	Hypothetical placeholder
124	E124	hypothetical element 124	Hypothetical placeholder
125	E125	hypothetical element 125	Hypothetical placeholder
126	E126	hypothetical element 126	Hypothetical placeholder
127	E127	hypothetical element 127	Hypothetical placeholder
128	E128	hypothetical element 128	Hypothetical placeholder
129	E129	hypothetical element 129	Hypothetical placeholder
130	E130	hypothetical element 130	Hypothetical placeholder
131	E131	hypothetical element 131	Hypothetical placeholder
132	E132	hypothetical element 132	Hypothetical placeholder
133	E133	hypothetical element 133	Hypothetical placeholder
134	E134	hypothetical element 134	Hypothetical placeholder
135	E135	hypothetical element 135	Hypothetical placeholder
136	E136	hypothetical element 136	Hypothetical placeholder
137	E137	hypothetical element 137	Hypothetical placeholder
138	E138	hypothetical element 138	Hypothetical placeholder

Particle category register

Six quark flavor categories, three charged-lepton particle/antiparticle categories, three neutrino flavor categories, and five boson categories. W⁺ and W⁻ are grouped; gluon color components are grouped. This is a table convention, not an absolute count of physical particles or quantum states.

Flavor labels differ from neutrino mass eigenstates. The register does not assert that the minimal massless-neutrino Standard Model explains observed neutrino masses, or resolve Dirac versus Majorana nature.

Sources: [CERN Standard Model](#) and [PDG 2026 tables](#). The 17 rows are categories under this convention, not all particle states or material ingredients.

Category	Labels	Accounting and status qualification
up quark flavor	u, anti-u	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
down quark flavor	d, anti-d	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
charm quark flavor	c, anti-c	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
strange quark flavor	s, anti-s	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
top quark flavor	t, anti-t	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
bottom quark flavor	b, anti-b	One flavor category; antiquark and three color states are not additional rows. No isolated free-quark ingredient is asserted.
electron and positron	e-, e+	Particle and antiparticle grouped in one category.
muon and antimuon	mu-, mu+	Particle and antiparticle grouped in one category.
tau and antitau	tau-, tau+	Particle and antiparticle grouped in one category.
electron-flavor neutrino	nu_electron, anti-nu_electron	Weak-interaction flavor category, not a mass eigenstate. Labels for neutrino/antineutrino interactions do not settle Dirac versus Majorana nature.
muon-flavor neutrino	nu_muon, anti-nu_muon	Weak-interaction flavor category, not a mass eigenstate. Labels for neutrino/antineutrino interactions do not settle Dirac versus Majorana nature.
tau-flavor neutrino	nu_tau, anti-nu_tau	Weak-interaction flavor category, not a mass eigenstate. Labels for neutrino/antineutrino interactions do not settle Dirac versus Majorana nature.
photon	gamma	One species category; momentum and polarization states are not separately enumerated.
gluon	g	Eight color components grouped in one category; no free isolated-gluon ingredient is asserted.
charged W bosons	W+, W-	Two charge-conjugate states grouped in one category.
Z boson	Z0	One species category; polarization states are not separate entries.
Higgs boson	H	Observed Higgs-boson category; additional hypothetical Higgs states are not asserted.

Composite particles, nuclear states, quasiparticles and hypothetical particle classes use the separate parameters in paragraphs [0046]-[0056]. Their possible identifiers are not additional recognized elemental rows. No complete current nuclide table or exhaustive theoretical-particle universe is claimed.

19 Claim support and review map

Textual support and physical enablement are different inquiries. Every claim remains novelty-unassessed. The complete per-claim map is supplied in the companion claim_support_map.json; the operative definitions and claims are printed here and do not depend on that external file.

Claims	Textual locations	Subject matter	Enablement and review status
1-138	[0010]-[0022], [0073]	Broad normalized material inventory and every exact component count.	Mathematical definitions present; composition-specific full-scope preparation absent. Counts above 118 necessarily include hypothetical labels.
139-147	[0010]-[0016], [0041], [0047]	Unary material, vocabulary restrictions and composition bases.	Identity or basis definitions do not establish novelty, availability or making methods.
148-168	[0017]-[0032], [0069]-[0072]	Rational and grid fractions, all endpoint ranges, complements and explicit boundary targets.	Exact arithmetic present; physical realization, tolerances and phase identity unresolved. Below-grid targets are broader than the floor-limited verifier.
169-173	[0046]-[0050]	Isotope, nuclear-state and time-indexed inventory descriptions.	Parameter identifiers do not establish nuclear existence, isolation, lifetime or synthesis.
174-185	[0012]-[0013], [0033]-[0043]	Structures, phases, spatial forms and measurement conventions.	Structural fields are specified as requirements, not completed material teachings for the universal composition domain.
186-193	[0005]-[0007], [0023]-[0032], [0073]-[0075]	Candidate-record construction and mathematical enumeration/feasibility methods.	Operations are mathematically described; novelty, eligibility, inventive contribution and full legal support remain unassessed.
194-196	[0051]-[0056]	Separate particle-system record methods.	Recording method does not prepare or stabilize every particle or hypothetical system.
197-199	[0005], [0041]-[0042], [0057]-[0068], [0075]	Conditional property scenarios and local evidence status.	No measured applicant result, inherent property or universal predicted performance is supplied.

20 Source and version notes

Source links identify attributed teachings, definitions, and evaluation editions. Mathematical models, record designs, numerical examples, and proposed scenarios are this draft's calculations or proposals. Source retrieval and local document creation dates are not patent filing or public-availability dates. No applicant identity, inventor signature, priority claim, micro-entity certification, complete current isotope archive, or worldwide patent-status finding is assumed. The later filing memorandum addresses procedure separately from this technical description.